

03EDA_PDF

February 6, 2026

```
[1]: # import packages
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path

# Load the Dataset
PROJECT_DIR = Path.cwd().resolve().parent
DATA_PATH = PROJECT_DIR / "data" / "processed" / "matchupDiff.csv"

assert DATA_PATH.exists(), f"Missing file: {DATA_PATH}"

matchupDiff = pd.read_csv(DATA_PATH)
matchupDiff.head()
```

```
[1]: Unnamed: 0  Season  DayNum  WTeamID  LTeamID  WScore  LScore  team1Id  \
0           0    2003     134     1421     1411      92      84     1411
1           1    2003     136     1112     1436      80      51     1112
2           2    2003     136     1113     1272      84      71     1113
3           3    2003     136     1141     1166      79      73     1141
4           4    2003     136     1143     1301      76      74     1143

      team2Id  y  ...  Raw Defensive Efficiency Rank_diff  \
0         1421  0  ...                               -165
1         1436  1  ...                               -34
2         1272  0  ...                               123
3         1166  0  ...                               159
4         1301  1  ...                               -39

      Raw Offensive Efficiency_diff Raw Offensive Efficiency Rank_diff  \
0                               1.6                               -43
1                               9.0                               -133
2                               3.2                               -42
3                               -5.4                                34
4                               -1.7                                19

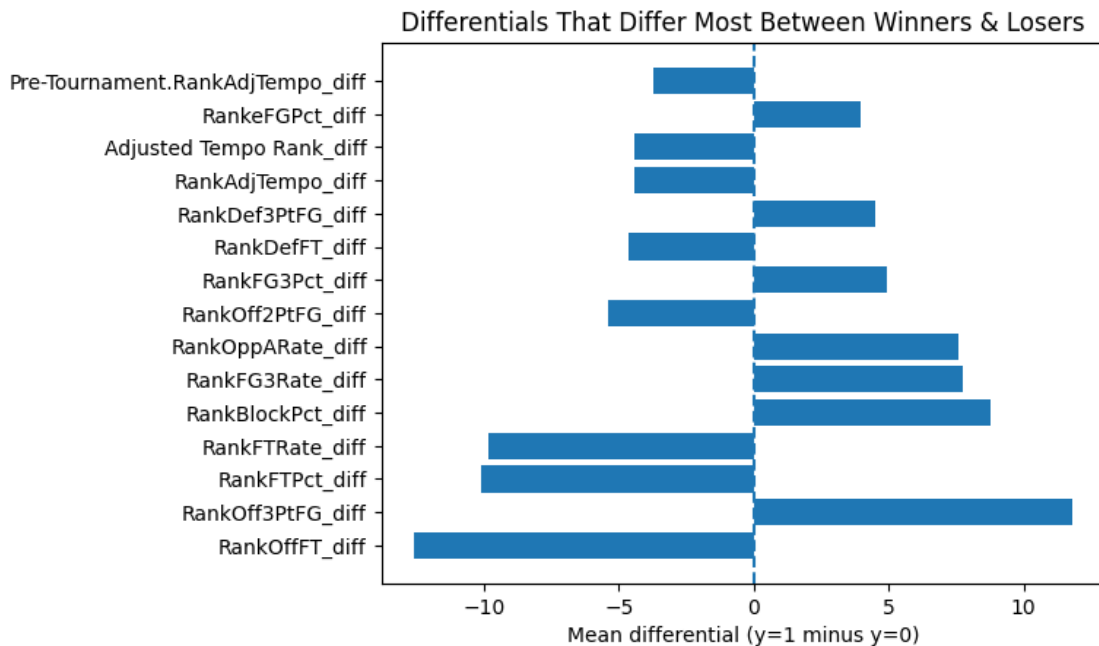
      Raw Tempo_diff  Raw Tempo Rank_diff  Seed_diff  StlRate_diff  TOPct_diff  \
0                1.1                -36            0        -0.0095        -1.0055
```

1	10.0	-263	-15	0.0061	-2.4229
2	-0.7	25	3	-0.0261	0.1265
3	3.3	-100	5	-0.0200	5.7503
4	2.2	-96	-1	-0.0206	-1.3144

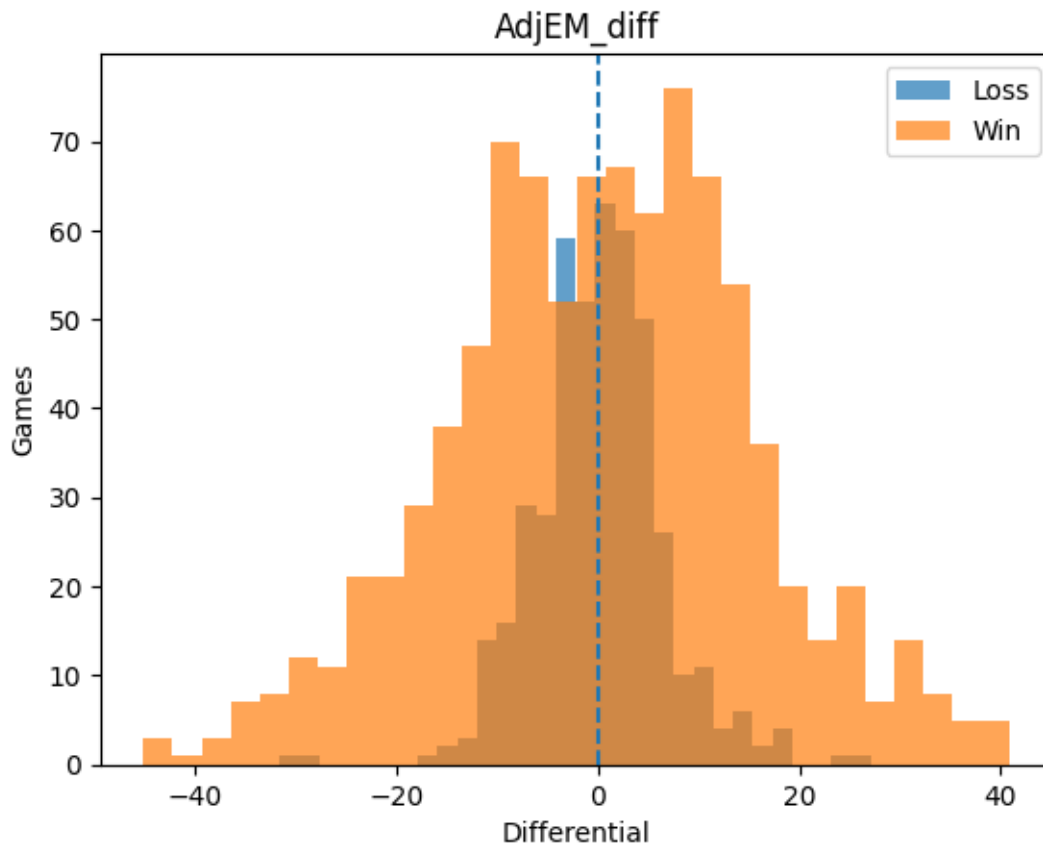
	Tempo_diff	eFGPct_diff
0	1.1023	1.6965
1	9.9893	2.7429
2	-0.6498	1.9572
3	3.3585	-0.0999
4	2.1439	-1.0445

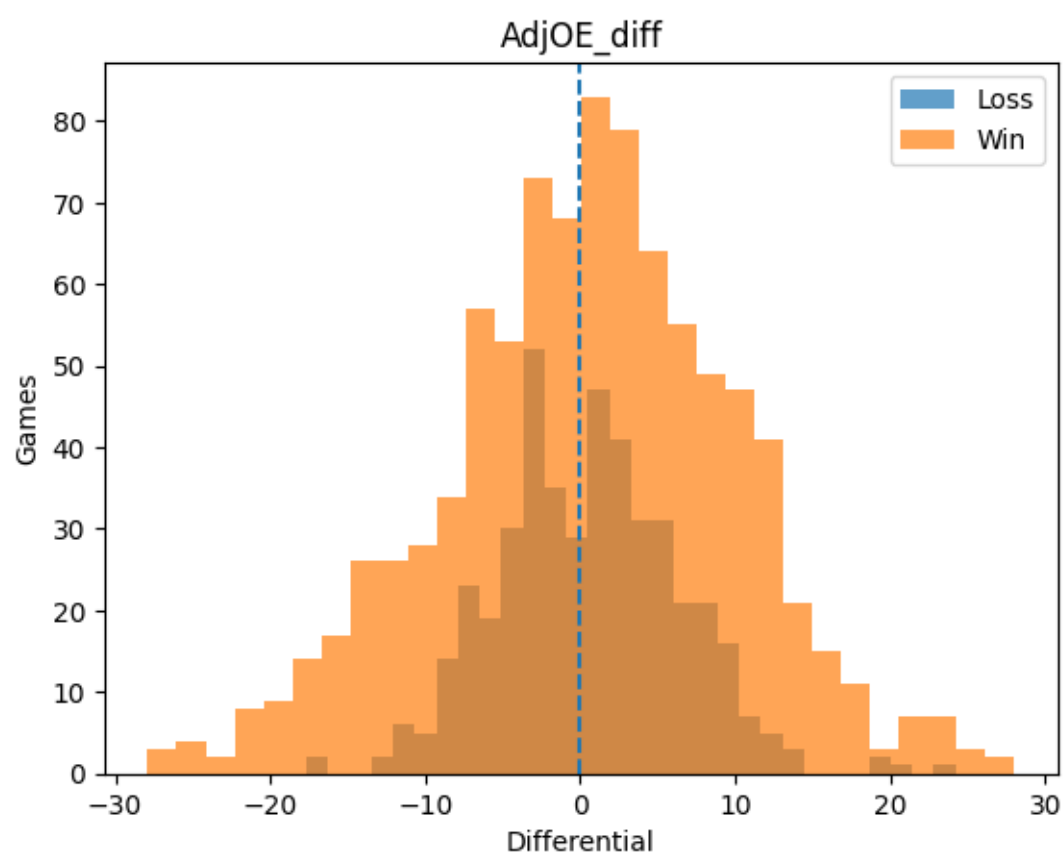
[5 rows x 105 columns]

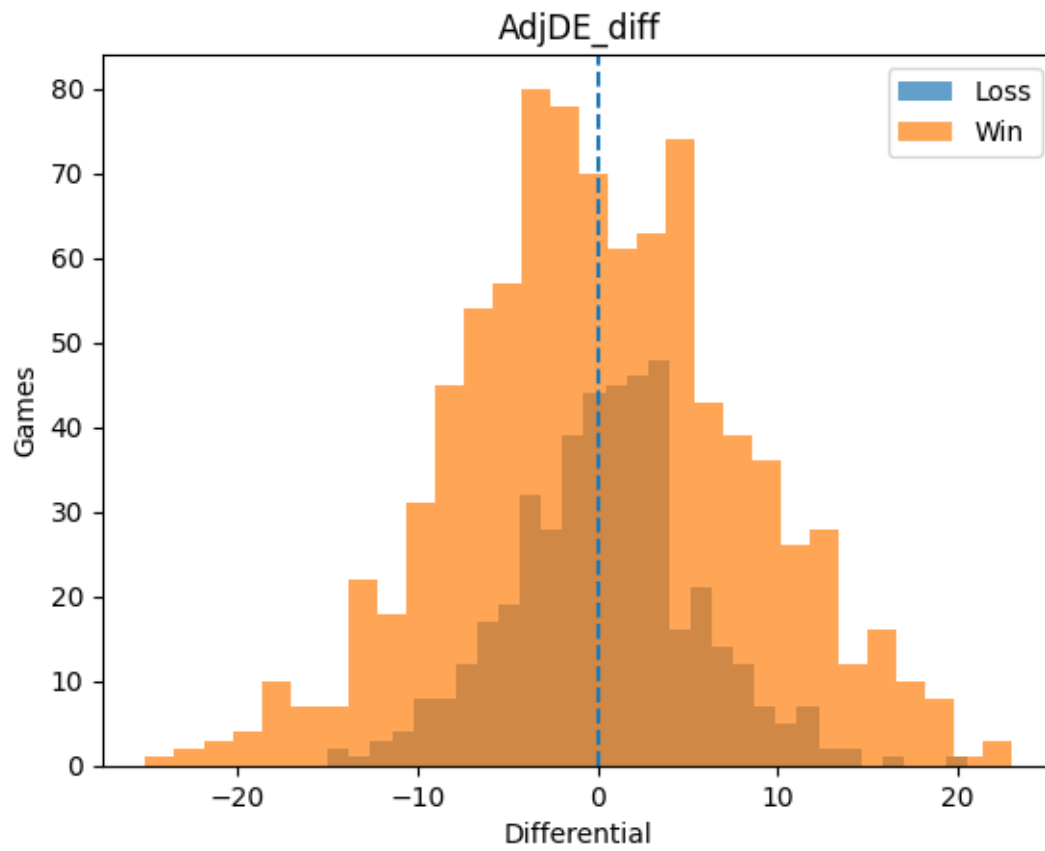
```
[3]: #1. Highest Differentials Between Winners and Losers
diff_cols = [c for c in matchupDiff.columns if c.endswith("_diff")]
means = matchupDiff.groupby("y")[diff_cols].mean()
mean_diff = (means.loc[1] - means.loc[0]).sort_values()
top = mean_diff.abs().sort_values(ascending=False).head(15).index
plt.barh(top, mean_diff[top])
plt.axvline(0, linestyle="--")
plt.title("Differentials That Differ Most Between Winners & Losers")
plt.xlabel("Mean differential (y=1 minus y=0)")
plt.show()
```

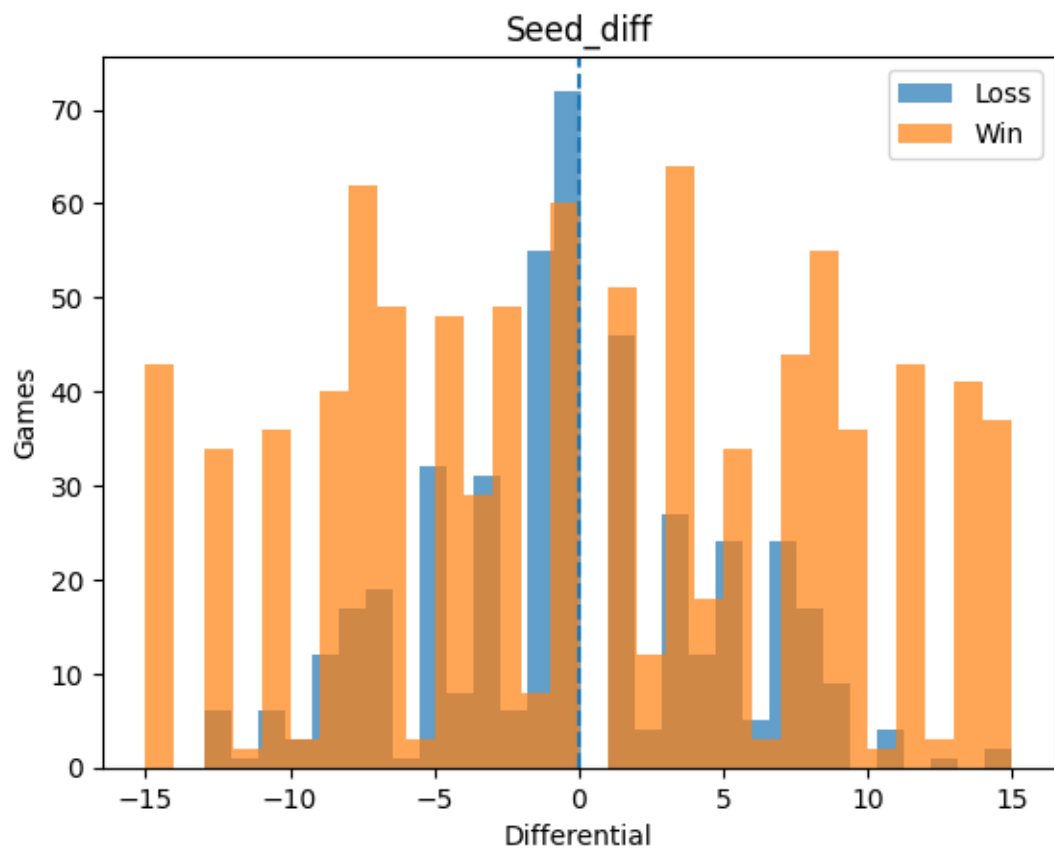


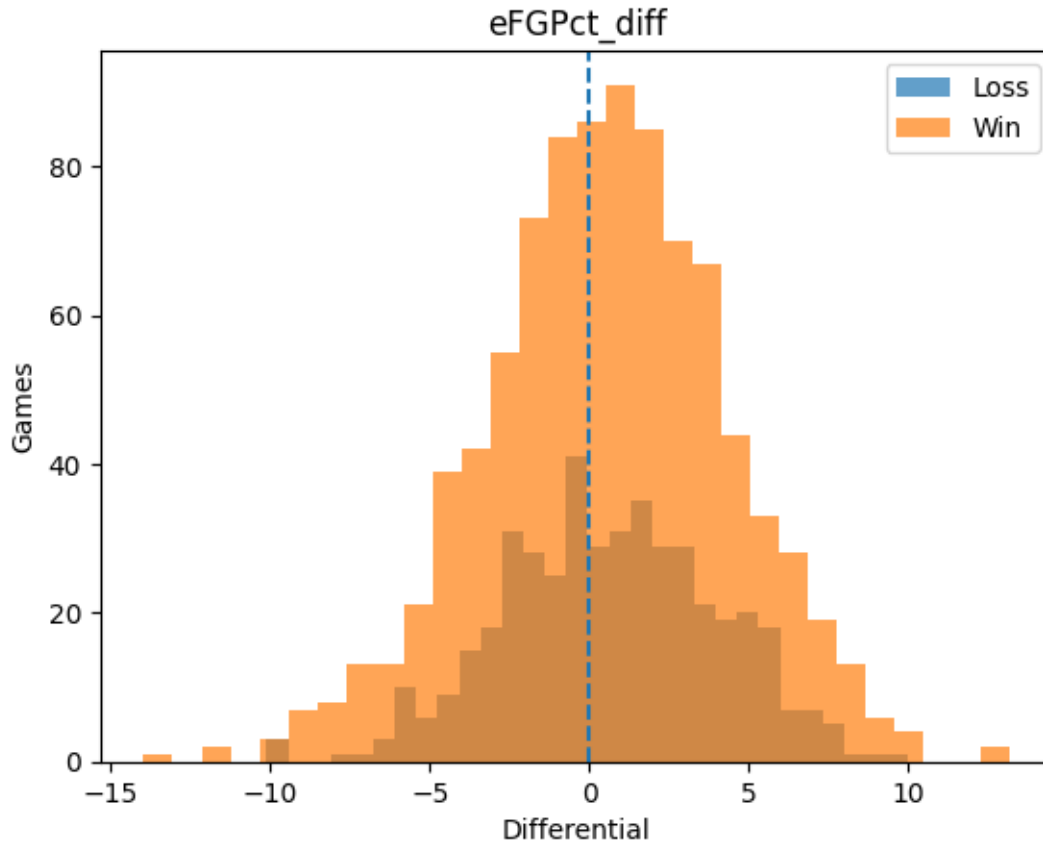
```
[4]: #2. Distributions of Winners and Losers for Key Statistics
features = [
    "AdjEM_diff",
    "AdjOE_diff",
    "AdjDE_diff",
    "Seed_diff",
    "eFGPct_diff"
]
for col in features:
    plt.figure()
    plt.hist(matchupDiff[matchupDiff["y"]==0][col], bins=30, alpha=0.7,
    ↪label="Loss")
    plt.hist(matchupDiff[matchupDiff["y"]==1][col], bins=30, alpha=0.7,
    ↪label="Win")
    plt.axvline(0, linestyle="--")
    plt.title(col)
    plt.xlabel("Differential")
    plt.ylabel("Games")
    plt.legend()
    plt.show()
```



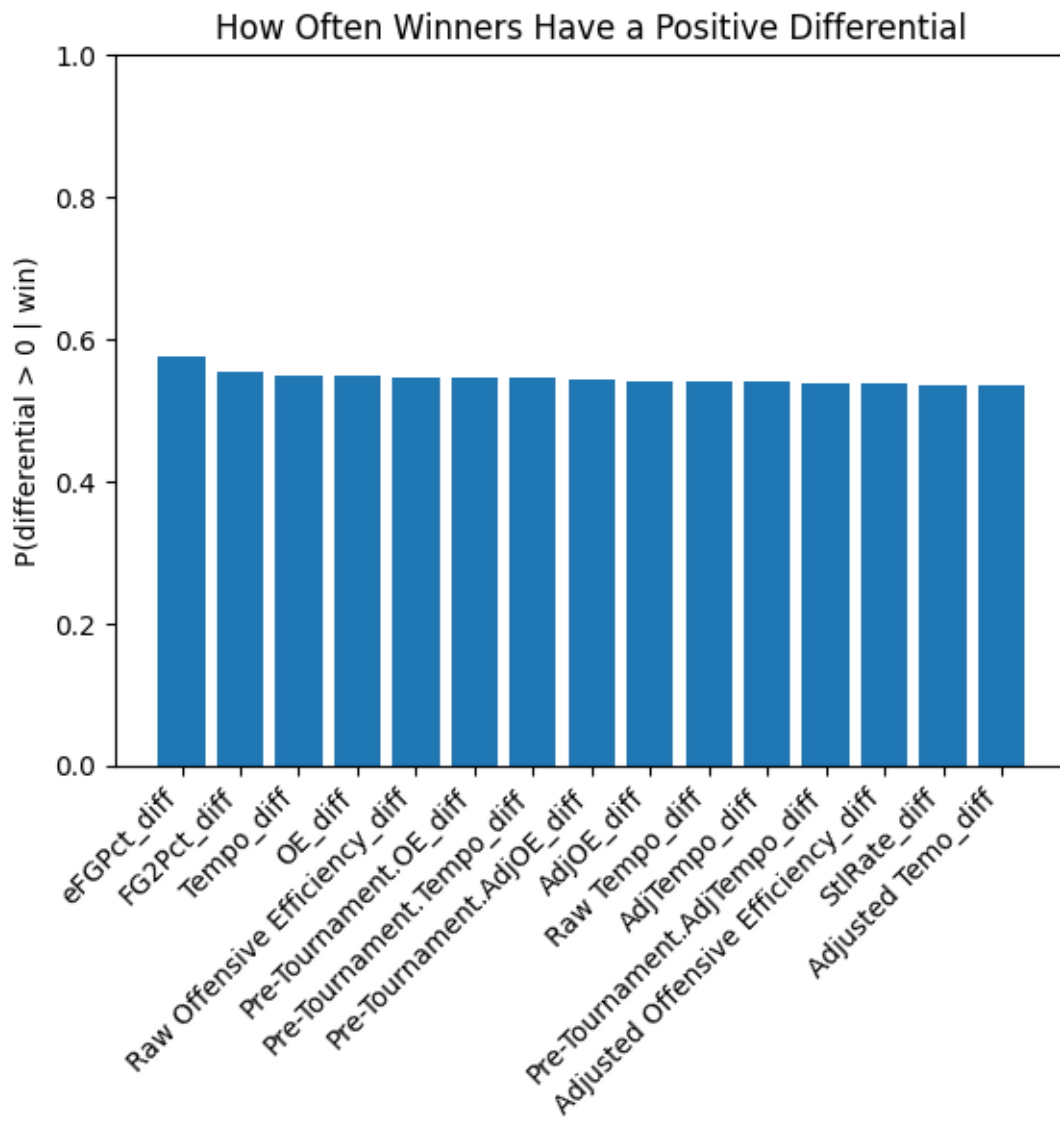




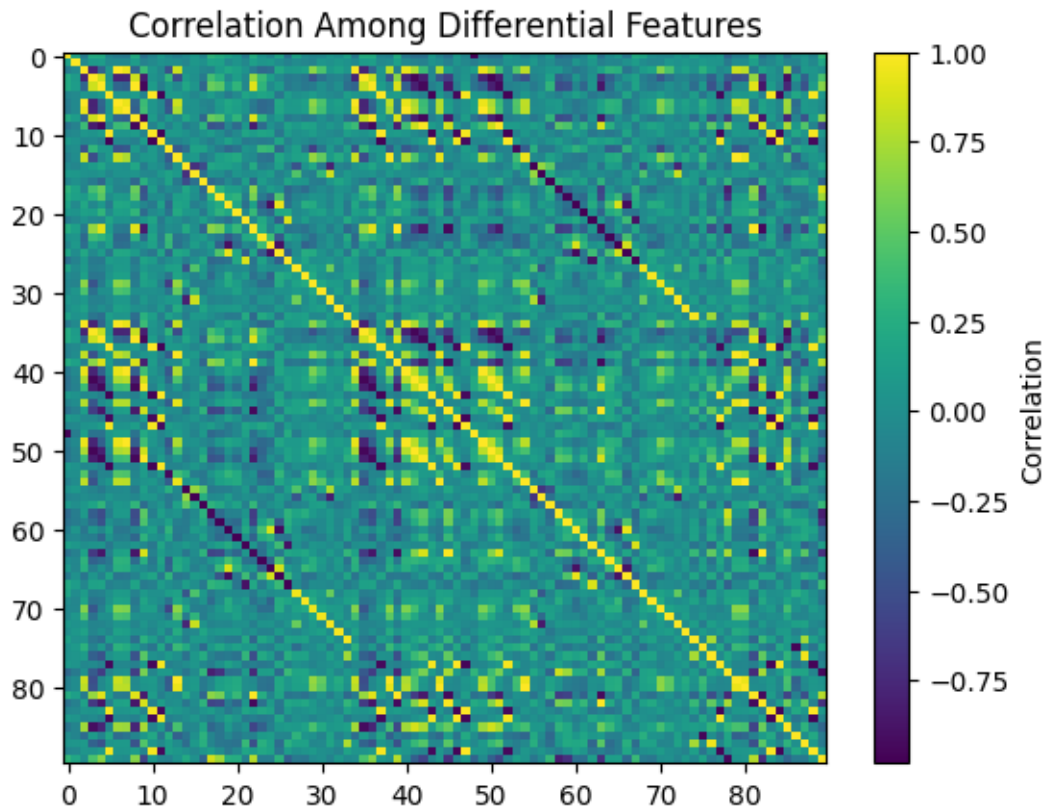




```
[5]: #3. Percentage that Winners Have a Positive Differential
pos_rate = {
    col: (matchupDiff[matchupDiff["y"] == 1][col] > 0).mean()
    for col in diff_cols
}
pos_rate = pd.Series(pos_rate).sort_values(ascending=False).head(15)
plt.bar(pos_rate.index, pos_rate.values)
plt.xticks(rotation=45, ha="right")
plt.ylim(0,1)
plt.ylabel("P(differential > 0 | win)")
plt.title("How Often Winners Have a Positive Differential")
plt.show()
```



```
[6]: #4. Correlation Heatmap for Differential Features
corr = matchupDiff[diff_cols].corr()
plt.imshow(corr, aspect="auto")
plt.colorbar(label="Correlation")
plt.title("Correlation Among Differential Features")
plt.show()
```

```
[7]: #5. Distribution of Seed Differentials for Wins and Losses
plt.figure()
plt.boxplot(
    [matchupDiff[matchupDiff["y"] == 0]["Seed_diff"].dropna(),
     matchupDiff[matchupDiff["y"] == 1]["Seed_diff"].dropna()],
    tick_labels=["Loss", "Win"]
)
plt.axhline(0, linestyle="--")
plt.ylabel("Seed differential")
plt.title("Seed Differential by Outcome")
plt.show()
```

